

1)

Claim:

Running time $T(n,x)$ is $\Theta(g(n))$ for all input x of size n iff the worst case $T_{\max}(n) = \max_x T(n,x)$ is $O(g(n))$ and the best case $T_{\min}(n) = \min_x T(n,x)$ is $\Omega(g(n))$

Proof:

suppose $T(n,x) = \Theta(g(n))$ for all inputs x .

Θ means $c_1, c_2 > 0$ and n_0 , for all $n \geq n_0$ and all inputs x

$$c_1 g(n) \leq T(n,x) \leq c_2 g(n)$$

minimum over x gives $T_{\min}(n) \geq c_1 g(n)$ for all $n \geq n_0$,
so $T_{\min}(n) = \Omega(g(n))$

maximum over x gives $T_{\max}(n) \leq c_2 g(n)$ for all $n \geq n_0$,
so $T_{\max}(n) = O(g(n))$

Suppose $T_{\max}(n) = O(g(n))$ and $T_{\min}(n) = \Omega(g(n))$

then $C, c > 0$ and N_1, N_2 for all $n \geq N_1$

$$T_{\max}(n) \leq Cg(n),$$

for all $n \geq N_2$

$$T_{\min}(n) \geq cg(n)$$

Let $N = \max(N_1, N_2)$, for any $n \geq N$ and any input x ,

$$cg(n) \leq T_{\min}(n) \leq T(n,x) \leq T_{\max}(n) \leq Cg(n)$$

so $cg(n) \leq T(n,x) \leq Cg(n)$ for all $n \geq N$ and all x .

$$T(n,x) = \Theta(g(n))$$

2)

proof:

limit $n \rightarrow \infty$ $\lg n / \sqrt{n}$

n is real variable x ,

do L Hopitals rule,

$$\begin{aligned} &= \lim_{x \rightarrow \infty} (1/(\ln 2)) * (1/x) / 1/2x^{-1/2} \\ &= \lim_{x \rightarrow \infty} 2 / \ln 2 * 1 / x^{1/2} = 0 \end{aligned}$$

since ratio go to 0, $\lg n = O(\sqrt{n})$,
so $\sqrt{n}$ grows faster than $\lg n$.

3)

Proof:

for all $n \geq 1$, $1/n \leq 1$,
so $e^{1/n} \leq e$.

for all $n \geq 1$, $n^t \geq 1$.

$C = e$ and $n_0 = 1$, so for all $n \geq n_0$,

$$e^{1/n} \leq e = C * 1 \leq C * n^t$$

So $e^{1/n} \leq Cn^t$ for all $n \geq n_0$,

$$e^{1/n} = O(n^t)$$

4)

a) $f(n) = 3n^2$, $g(n) = n^2$

$$f(n) = 3n^2 = 3 * g(n)$$

$$\text{so } f(n) = \Theta(g(n))$$

b) $f(n) = 2n^4 - 3n^2 + 7$, $g(n) = n^5$

for large n , $f(n)$ acts like $2n^4$ while $g(n) = n^5$,

So $f(n) = O(g(n))$, but $f(n)$ is not $\Omega(g(n))$

$f \in O(g)$ shows this

c) $f(n) = \log n$, $g(n) = \log n + 1/n$,

for large n , $1/n$ is compared to $\log n$,

So $g(n) \rightarrow \log n$.

theres not n_0 and $c_1, c_2, > 0$ for all $n \geq n_0$.

$$c_1(\log n + 1/n) \leq \log n \leq c_2(\log n + 1/n),$$

so $\log n = \Theta(\log n + 1/n)$,

$$f(n) = \Theta(g(n))$$

d) $f(n) = 2^{k \log n}$, $g(n) = n^k$

assume $\log$ is base 2, $2^{\log n} = n$,

$$\text{so } 2^{k \log n} = (2^{\log n})^k = n^k.$$

so $f(n) = g(n)$ and $F(n) = \Theta(g(n))$

e) $f(n) = 2^n$, $g(n) = 2^{2n} = 4^n$

$$f(n) / g(n) = 2^n / 4^n = (1/2)^n = 0$$

So $f(n) = O(g(n))$, but not $\Omega(g(n))$.

$f \in O(g)$ only

5)

proof:

$$\sqrt[n]{n+1}^3 = (n+1)^{3/2}$$

all $n \geq 0$ we have $n+1$, so $(n+1)^{3/2} \geq n^{3/2}$,

$$n^{3/2} = n \sqrt{n}.$$

for every $n \geq 0$

$$\sqrt[n]{n+1}^3 \geq 1 * n \sqrt{n}$$

constant $c = 1$ and $n_0 = 0$ shows by Ω ,

$$\sqrt[n]{n+1}^3 = \Omega(n \sqrt{n})$$